

Problem Set 4

Exercise 1 Using the data on 4,137 college students, the following equation was estimated by OLS:

$$\widehat{colgpa} = 1.392 - .0135 \text{ hspc} + .00148 \text{ sat}$$

where *colgpa* is measured on a four-point scale, *hsperc* is the percentile in the high school graduating class (defined so that, for example, *hsperc* = 5 means the top 5% of the class), and *sat* is the combined math and verbal scores on the student achievement test. The R^2 of the regression is .273.

- (a) Why does it make sense for the coefficient on *hsperc* to be negative?
- (b) What is the predicted college GPA when *hsperc* = 20 and *sat* = 1,050?
- (c) Suppose that two high school graduates, A and B, graduated in the same percentile from high school, but Student A's SAT score was 140 points higher (about one standard deviation in the sample). What is the predicted difference in college GPA for these two students? Is the difference large?
- (d) Holding *hsperc* fixed, what difference in SAT scores leads to a predicted *colgpa* difference of .50, or one-half of a grade point? Comment on your answer.

Exercise 2 The following model is a simplified version of the multiple regression model used by Biddle and Hamermesh (1990) to study the tradeoff between time spent sleeping and working and to look at other factors affecting sleep:

$$\text{sleep} = \beta_0 + \beta_1 \text{totwrk} + \beta_2 \text{educ} + \beta_3 \text{age} + u$$

where *sleep* and *totwrk* (total work) are measured in minutes per week and *educ* and *age* are measured in years.

- (a) If adults trade off sleep for work, what is the sign of β_1 ?
- (b) What signs do you think β_2 and β_3 will have?
- (c) The estimated coefficients ($n = 706$) read

$$\widehat{\text{sleep}} = 3,638.25 - .148 \text{ totwrk} - 11.13 \text{ educ} + 2.20 \text{ age} + u$$

The R^2 of the regression is .113.

If someone works five more hours per week, by how many minutes is sleep predicted to fall? Is this a large tradeoff?

- (d) Discuss the sign and magnitude of the estimated coefficient on *educ*.

- (e) Would you say *totwrk*, *educ*, and *age* explain much of the variation in sleep? What other factors might affect the time spent sleeping? Are these likely to be correlated with *totwrk*?

Exercise 3 Suppose you are interested in estimating the *ceteris paribus* relationship between y and x_1 . For this purpose, you can collect data on two control variables, x_2 and x_3 . (For concreteness, you might think of y as the final exam score, x_1 as class attendance, x_2 as GPA up through the previous semester, and x_3 as SAT or ACT score.) Let $\tilde{\beta}_1$ be the simple regression estimate from y on x_1 and let $\hat{\beta}_1$ be the multiple regression estimate from y on x_1, x_2, x_3 .

- If x_1 is highly correlated with x_2 and x_3 in the sample, and x_2 and x_3 have large partial effects on y , would you expect $\tilde{\beta}_1$ and $\hat{\beta}_1$ to be similar or very different? Explain.
- If x_1 is almost uncorrelated with x_2 and x_3 , but x_2 and x_3 are highly correlated, will $\tilde{\beta}_1$ and $\hat{\beta}_1$ tend to be similar or very different? Explain.
- If x_1 is highly correlated with x_2 and x_3 , and x_2 and x_3 have small partial effects on y , would you expect $SE(\tilde{\beta}_1)$ or $SE(\hat{\beta}_1)$ to be smaller? Explain.
- If x_1 is almost uncorrelated with x_2 and x_3 , x_2 and x_3 have large partial effects on y , and x_2 and x_3 are highly correlated, would you expect $SE(\tilde{\beta}_1)$ or $SE(\hat{\beta}_1)$ to be smaller? Explain.

Exercise 4 Research indicates that low birthweight babies are at a higher risk for various health complications and developmental challenges, which can persist throughout their lives. Inadequate birthweight is often linked to poorer cognitive abilities and a higher likelihood of chronic illnesses, limiting educational, employment opportunities, and personal development.

You are asked to study the impact of smoking on birthweight. You access a dataset collected on a random sample of 3000 babies born in Pennsylvania in 1989 from the 1989 linked National Natality-Mortality Detail files. The data include the baby's birth weight together with various characteristics of the mother, including whether she smoked during the pregnancy. The average birthweight is 3383 grams, with a standard deviation of 592 grams. The smallest registered birthweight in your sample is 425 grams, and the largest is 5755 grams.

The variables are the following:

- *birthweight*: birth weight of infant (in grams)
- *smoker*: indicator equal to one if the mother smoked during pregnancy and zero, otherwise.
- *alcohol*: indicator=1 if mother drank alcohol during pregnancy
- *educ*: years of educational attainment (more than 16 years coded as 17)

- # prenatal visits: total number of prenatal visits
 - no prenatal visits: indicator=1 if no prenatal care visit was made
 - prenatal visits (1st semester): indicator=1 if 1st prenatal care visit in 1st trimester
 - prenatal visits (2nd semester): indicator=1 if 1st prenatal care visit in 2nd trimester
 - prenatal visits (3rd semester): indicator=1 if 1st prenatal care visit in 3rd trimester
-
- (a) Before collecting data, you are thinking about the ideal experiment to study how smoking affects birthweight. What would it be? Why would it be ideal? What are the practical limits to running such an ideal experiment?
 - (b) Consider column (1). What is the estimated effect of smoking on birthweight? Is the coefficient statistically significant? Is the magnitude of the coefficient large?
 - (c) Consider columns (2)-(4). Why would you include other variables such as alcohol or # prenatal visits in your regression specification?
 - (d) Jane smoked during her pregnancy, did not drink alcohol, and had 3 prenatal care visits. Use the regression in column (3), to approximately predict the birth weight of Jane's child.
 - (e) Consider columns (3)-(4). How should you interpret the coefficient on # prenatal visits? Does the coefficients measure a causal effect of prenatal visits on birth weight? If not, what does it measure?
 - (f) Consider column (5). An alternative way to control for prenatal visits is to use the binary variables prenatal visits (1st semester), prenatal visits (2nd semester), prenatal visits (3rd semester) and no prenatal visits. Why is prenatal visits (3rd semester) excluded from the regression?
 - (g) Does the regression in column (5) explain a larger fraction of the variance in birthweight than the regression in column (4)?
 - (h) Consider columns (1)-(5). Have you identified the causal effect of smoking on birthweight? Discuss.

	<i>Dependent Variable: Birthweight</i>				
	(1)	(2)	(3)	(4)	(5)
<i>smoker</i>	-253.228*** (26.81)	-250.803*** (26.87)	-217.580*** (26.11)	-207.932*** (27.12)	-213.431*** (27.58)
<i>alcohol</i>		-57.601 (77.39)	-30.491 (72.60)	-35.491 (72.76)	-23.942 (69.94)
<i># prenatal visits</i>			34.070*** (3.61)	33.168*** (3.64)	
<i>years of education</i>				8.118 (5.01)	13.724** (5.17)
<i>no prenatal visits</i>					-563.298*** (160.70)
<i>prenatal visits (1st semester)</i>					117.176 (67.39)
<i>prenatal visits (2nd semester)</i>					34.532 (72.41)
<i>prenatal visits (3rd semester)</i>					omitted (.)
<i>constant</i>	3432.060*** (11.89)	3432.703*** (11.94)	3051.249*** (43.71)	2954.604*** (74.79)	3153.803*** (93.27)
<i>N</i>	3000.000	3000.000	3000.000	3000.000	3000.000
<i>F – statistic</i>	89.211	44.754	59.484	45.578	21.146
<i>R²</i>	0.029	0.029	0.073	0.074	0.049

Robust standard errors are reported in parenthesis. *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$.

Table 1: Effect of *Smoker* on *Birthweight*.

Exercise 5 Answer the following questions:

- What is a joint hypothesis?
- Explain how an F -statistic is constructed to test a joint hypothesis.
- What is the hypothesis that is tested by constructing the overall regression F -statistic in the multiple regression model $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$? Explain using the concepts of restricted and unrestricted regressions.
- Why is it important for a researcher to have information on the distribution of the error terms when implementing these tests?